

## CIS170 Project 7

Wibaux University's HR department wants an instructor report that shows the number of years until each is eligible for retirement. The instructor is eligible for retirement after being employed for 30 years. The data should be presented in alphabetical order by last name and then first name; this will require a logical file (or an SQL index) over the Wibaux University instructor's file (WUINSTP) sorted by the instructor's last name and then by first name.

WUINSTP Input:

| A |           |      | UNIQUE                    |
|---|-----------|------|---------------------------|
| A | R INSTREC |      |                           |
| A | INSTNO    | 9S 0 | TEXT('Instructor SSN')    |
| A | IFNAME    | 10   | TEXT('First Name')        |
| A | ILNAME    | 15   | TEXT('Last Name')         |
| A | DEPT      | 3    | TEXT('Course Department') |
| A | SALARY    | 8S 2 | TEXT('Salary')            |
| A | RANK      | 1    | TEXT('Academic Rank')     |
| A | SEX       | 1    | TEXT('Sex')               |
| A | HIRDAT    | 8S 0 | TEXT('Date of Hire')      |
| A | MARSTS    | 1    | TEXT('Marital Status')    |
| A | DEPEND    | 2S 0 | TEXT('Dependents')        |
| A | TENURE    | 1    | TEXT('Tenured')           |
| A | TITLE     | 1    | TEXT('Preferred Title')   |
| A | STREET    | 20   | TEXT('Street Address')    |
| A | CITY      | 15   | TEXT('City')              |
| A | STATE     | 2    | TEXT('State')             |
| A | ZIP       | 9S 0 | TEXT('Zip')               |
| A | IEMAIL    | 30   | TEXT('Instructor Email')  |
| A | K INSTNO  |      |                           |

| Wibaux University            |                                  |              |      |               |        |              |            |            |  |
|------------------------------|----------------------------------|--------------|------|---------------|--------|--------------|------------|------------|--|
| Instructor Retirement Report |                                  |              |      |               |        |              |            |            |  |
| Instructor                   |                                  | Date         |      | Eligible      |        | Years Before |            |            |  |
| Number                       | Instructor Name                  | Dept.        | Rank | Current Wage  | Tenure | Hired        | Date       | Retirement |  |
| 666666666                    | 00000000000000000000000000000000 | 000          | 0    | \$666,666.66- | 0      | 66/66/6666   | mm/dd/yyyy | 66         |  |
| Average Wage:                |                                  | 666,666,666- |      |               |        |              |            |            |  |
| Total Instructors:           |                                  | 6,666        |      |               |        |              |            |            |  |

Report: RETIREREP – NOTE – ALREADY CREATED – SEE PROGRAMMING NOTES ON NEXT PAGE

--- continued on next page ---

## **Project Approach:**

For this project, some of the coding has already been completed. In a team coding environment, different tasks might be assigned to different programmers, such as creating the external printer definition. Or, perhaps a logical file definition already exists, so you reuse it. Of course, the programmer using pre-existing code will need to become familiar with the field names, etc. already in use in these instances. This project simulates some of those scenarios by providing parts of the coding.

## **Notes:**

1. The external logical file INSTRUCTLF has been placed in your QDDSSRC file. Because the list is to be printed in alphabetical order and the original key to the physical file is the instructor number, a logical file is needed. (This is a topic covered in CIS121.) Open INSTRUCTLF, review its code, and compile it so that it can be used in your program.
2. The external printer definition RETIREREP has been created and has been placed in your QDDSSRC file. Open it and read it carefully, it contains notes for your reference. Compile it so that you can use it in your program.

## **Programming Notes:**

1. Your input file will be INSTRUCTLF and your output will be RETIREREP (see above)
  - a. In order to actually utilize the new keyed order of the logical file, we have to add the keyword "Keyed" to our DCL-F for the INSTRUCTLF file. This makes it use the keyed order of the file instead of "arrival sequence" for the file. See page 259 for an example of using this keyword.
2. HIRDAT is in the file as a "legacy date" which has been defined as a simple number, in ccyyymmdd order, for example 20150203 (2015 is ccyy, 02 is mm, 03 is dd)  
  
Convert HIRDAT to a native date field called HIREDATE so that it can be used in calculations.
3. Calculate the retirement eligible date -- RDATE -- (thirty years from the hire date).
4. Calculate the years before retirement -- RYEARS --how many years from the system date(today)
5. Note that the instructors name is to be trimmed and concatenated and printed as one item with only one space between the first and last name. (using techniques from last project)
6. Calculate the average yearly wage for all instructors and print it along with the number of instructors processed. As you process each record, accumulate a grand total of salary and accumulate a count. After all the records have been processed, calculate the average and then print it along with the count.